

$$\textcircled{5} \quad \textcircled{1} \quad a = -\omega^2 x$$

$$\frac{-123}{0.1} = -\omega^2$$

$$\omega = 35.07 \text{ rad/sec}$$

$$\omega = 2\pi f$$

$$f = \frac{35.07}{2\pi} = 5.58 \text{ Hz}$$

$$x = 0.1 \text{ m}$$

$$a = -123 \text{ m/s}^2$$

$$k = 400 \text{ N/m}$$

$$v = 13.6 \text{ m/s}$$

$$\textcircled{11} \quad \omega^2 = \frac{k}{m}$$

$$m = \frac{k}{\omega^2} = \frac{400}{(35.07)^2} = 0.325 \text{ kg}$$

$$\textcircled{111} \quad v = \omega \sqrt{A^2 - x^2}$$

$$\sqrt{A^2 - x^2} = \frac{v}{\omega}$$

$$A^2 = \frac{v^2}{\omega^2} + x^2$$

$$A^2 = \frac{(13.6)^2}{(35.07)^2} + (0.1)^2$$

$$A = 0.4 \text{ m (Ans)}$$

$$(vi) E_T = 0.0315 + 0.04 = 0.0715$$

$$E_T = \frac{1}{2} K A^2$$

$$A^2 = \frac{2 E_T}{K}$$

$$A^2 = \frac{2 \times 0.0715}{70}$$

$$A = 0.045$$

$$\boxed{14} \quad \textcircled{1} \quad \omega = \frac{106}{2}$$

$$\omega = 9.899$$

$$f = 2\pi \times 9.899$$

$$= 62.2 \text{ Hz}$$

$$T = 0.016 \text{ sec}$$

$$m = 2 \text{ kg}$$

$$k = 106 \text{ N/m}$$

$$x = 5 \text{ cm}$$

$$= 0.05 \text{ m}$$

$$\textcircled{11} \quad x = A \sin 9.899t$$

Q

(11)

$$\omega = \frac{k}{m}$$

$$= \frac{0.4}{0.025}$$

$$\therefore \omega = 4,$$

$$\omega = 2\pi f$$

$$\therefore f = \frac{4}{2\pi} = 0.637 \text{ Hz}$$

$$\omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57 \text{ sec}$$

(12)

$$x = A \sin \omega t$$

$$0.1 = A \sin 4t$$

$$m = 25 \text{ gm} = 0.025 \text{ kg}$$

$$k = 400 \text{ dyne/cm}$$

$$= 0.4 \text{ Nm}$$

$$x = 10 \text{ cm} = 0.1 \text{ m}$$

$$a_{\max} = -A\omega^2 = -10.52 \text{ ms}^{-2}$$

$$\textcircled{\text{VI}} \quad x = A \sin \omega t$$

$$x_{(0.5\text{sec})} = 0.11 \sin(0.78 \times 0) = 0 \text{ m}$$

$$x_{(7\text{sec})} = 0.11 \sin(0.78 \times 7) = 0.1023 \text{ m}$$

$$\textcircled{\text{VII}} \quad v = \omega \sqrt{A^2 - x^2}$$

$$0.22\text{sec} \rightarrow v = 0.78 \sqrt{(0.11)^2 - (0.11)^2}$$

$$\left. \begin{array}{l} m = 0.11 \\ m = 0.04 \end{array} \right\}$$

$$v = 0$$

$$(0.04\text{sec}) v = 0.78 \sqrt{(0.11)^2 - (0.04)^2}$$

$$= 1.0021 \text{ ms}^{-1}$$

$$\underline{21} \quad \omega^N = \frac{k}{m}$$

$$\textcircled{11} \quad \omega^N = \frac{65}{0.68}$$

$$\omega = \sqrt{\frac{65}{0.68}} = 9.78$$

$$\left| \begin{array}{l} m = 680 \text{ gm} = 0.68 \text{ kg} \\ k = 65 \text{ N/m} \\ A = 11 \text{ cm} = 0.11 \text{ m} \end{array} \right.$$

$$\textcircled{1} \quad \omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{9.78} = 0.642 \text{ sec}$$

$$\textcircled{1V} \quad V = -A\omega \sin(\omega t + \phi)$$

$$= -0.11 \times 9.78 \sin(9.78 t)$$

$$\begin{aligned} V(4\text{sec}) &= -1.0758 \sin(9.78 \times 4) \\ &= -0.679 \text{ m s}^{-1} \end{aligned}$$

$$V_{\max} = -A\omega = -0.11 \times 9.78 = -1.0758 \text{ m s}^{-1}$$

$$\textcircled{2V} \quad a = -A\omega^2 \cos \omega t$$

$$a(4\text{sec}) = -0.11 \times (9.78)^2 \cos(9.78 \times 4)$$

$$\begin{aligned} &= -10.52 \times 0.78 \\ &= -8.161 \text{ m s}^{-2} \end{aligned}$$

$$\boxed{3} \quad \textcircled{1} \quad F = kx$$

$$= 118.44 \times 0.085$$

$$= 9.974$$

$$\textcircled{11} \quad \omega^2 = \frac{k}{m}$$

$$\left(\frac{2\pi}{T}\right)^2 = \frac{k}{0.12}$$

$$k = 118.44$$

$$m = 0.12 \text{ kg}$$

$$A = 8.5 \text{ cm}$$

$$= 0.085 \text{ m}$$

$$T = 0.2 \text{ sec}$$

$$\boxed{4} \quad \textcircled{1} \quad x = 2 \text{ mm} = 2 \times 10^{-3} = A$$

$$\textcircled{11} \quad V = -A\omega$$

$$= -(2 \times 10^{-3}) \times (2\pi f)$$

$$= -(2 \times 10^{-3}) \times 2 \times \pi \times 120$$

$$= -1.50 \text{ m s}^{-1}$$

$$\textcircled{11} \quad a_{\max} = -A\omega^2$$

$$= -(2 \times 10^{-3}) \times (2\pi f)^2$$

$$= -1126.98 \text{ m s}^{-2}$$

$$x = 2 \text{ mm}$$

$$= 2 \times 10^{-3} \text{ m}$$

$$f = 120 \text{ Hz}$$

Ex 10

$$\lambda = 0.377 \text{ cm} = 0.00377 \text{ m}$$

$$f = 0.25 \text{ Hz}$$

(iii) $A = \lambda = 0.00377 \text{ m}$ [$\because v = \phi$]

$$\begin{aligned} \textcircled{i} \quad T &= \frac{1}{f} \\ &= \frac{1}{0.25} \\ &= 4 \text{ sec} \end{aligned}$$

$$\begin{aligned} \textcircled{ii} \quad \omega &= 2\pi f \\ &= 2\pi \times 0.25 \\ &= 1.57 \end{aligned}$$

$$\textcircled{iv} \quad x = A \sin \omega t$$

$$\begin{aligned} x_{(3\text{sec})} &= 0.00377 \sin(1.57 \times 3) \\ &= 3.038 \times 10^{-4} \text{ m} \end{aligned}$$

$$\begin{aligned} \textcircled{v} \quad v &= -A\omega \sin \omega t \\ &= -0.00377 \times 1.57 \sin(1.57 \times 3) \\ &= -4.769 \times 10^{-4} \text{ m} \end{aligned}$$

$$\boxed{22} \quad x = 10 \cos\left(3\pi t + \frac{\pi}{3}\right)$$

$$\textcircled{1} \quad x(2.5) = 10 \cos\left[\left(3\pi \times 2.5\right) + \frac{\pi}{3}\right]$$

$$\boxed{13} \quad \textcircled{1} \quad T = \frac{1}{f} = \frac{1}{0.5} = 2 \text{ sec} \quad \left| \begin{array}{l} A = 4 \text{ m} \\ f = 0.5 \text{ Hz} \\ \phi = \frac{\pi}{4} \end{array} \right.$$

$$\textcircled{11} \quad x = 4 \sin\left(\pi t + \frac{\pi}{4}\right) \quad (Am) \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$\textcircled{14} \quad V(5s) = -A\omega \sin(\omega t + \phi)$$

$$= -4 \times \pi \sin\left(5\pi + \frac{\pi}{4}\right)$$

$$a(5s) = -A\omega^2 \cos(\omega t + \phi) = -4\pi^2 \cos\left(\pi t + \frac{\pi}{4}\right)$$

15 (i)

$$F = -kx$$

$$mg = -kx$$

$$k = -\frac{0.5 \times 9.8}{0.07} = 70 \text{ N/m}$$

$$m = 500 \text{ gm}$$

$$= 0.5 \text{ Kg}$$

$$x = 7 \text{ cm} = 0.07 \text{ m}$$

$$(ii) \omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{70}{0.5}} = 11.83 \text{ rad/s}$$

$$(iii) T = \frac{2\pi}{\omega} = \frac{2\pi}{11.83} = 0.531 \text{ sec}$$

$$(iv) E_p = \frac{1}{2} kx^2$$

$$= \frac{1}{2} \times 70 \times (0.07)^2$$

$$= 0.1715 \text{ J}$$

$$x = A = 3 \text{ cm}$$

$$= 0.03 \text{ m}$$

$$V = 40 \text{ cm/s}$$

$$= 0.4 \text{ m/s}$$

$$\textcircled{\checkmark} E_k = \frac{1}{2} mv^2$$

$$= \frac{1}{2} \times 0.5 \times (0.4)^2$$

$$= 0.04 \text{ J}$$

$$\boxed{16} (a) \overset{(c)}{\omega^2} = \frac{k}{m}$$

$$= \frac{0.4}{0.025}$$

$$\omega = 4 \text{ rad/sec}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57 \text{ sec}$$

$$m = 25 \text{ gm}$$

$$= 0.025 \text{ kg}$$

$$k = 400 \text{ dyne/cm}$$

$$= 0.4 \text{ N/m}$$

$$x = 10 \text{ cm}$$

$$= 0.1 \text{ m}$$

$$(b) f = \frac{1}{T} = \frac{1}{1.57} = 0.637 \text{ Hz}$$

$$(d) E = \frac{1}{2} k A^2 =$$

